D. J. Rasmussen, Ph.D.

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RESEARCH INTERESTS

sea-level rise and coastal floods (physical science), coastal flood defense (social aspects), decision-making under uncertainty, natural hazard risk management, policy process for public works, environmental policy, climate change (physical science)

RECENT PROFESSIONAL EXPERIENCE

2022–now Climate Analytics Practitioner

2021–2022 Postdoctoral Research Associate, School of Public and International Affairs, Princeton University

EDUCATION

2021 Ph.D. in Public and International Affairs, Princeton University

Dissertation: *Coastal Defense in an Era of Sea-Level Rise: Science, Politics, and Decision-Making* [pdf]

Committee: *Michael Oppenheimer, Robert E. Kopp, Ning Lin, Guy Nordenson*

2013 M.S. in Civil and Environmental Engineering, University of California, Davis

Thesis: *The Ozone Climate Penalty: Past, Present, and Future* [pdf]

2009 B.S. in Atmospheric and Oceanic Science, University of Wisconsin, Madison

Thesis: *An Analysis of Climate Change Impacts on Future Wind Energy Production in California* [pdf]

OTHER PROFESSIONAL EXPERIENCE

2013–2016 Contracted Climate Risk Analyst, Rhodium Group

2014–2015 Senior Engineer, ENVIRON/ Ramboll

2012–2014 Research Assistant and then Engineer, California Air Resources Board

2010–2011 Research Analyst, Geophysical Fluid Dynamics Laboratory/ NOAA

2008–2010 Research Assistant, UW-Madison

JOURNAL ARTICLES

2026 **D.J. Rasmussen**. Multivariate bias correction of ERA5 using in-situ observations for planning and engineering. *Environmental Research: Climate*. **5** (2) 025026 doi:10.1088/2752-5295/ae63ee [pdf]

2025 J. Morim, **D.J. Rasmussen**, T. Wahl, F.M. Calafat, R.E. Kopp, and M. Oppenheimer. US-CoastEX: Observation-based probabilistic reanalysis of storm surge and sea level extremes for the United States. *Scientific Data* 12, 1395. doi:10.1038/s41597-025-05730-1 [pdf]

2025 J. Morim, T. Wahl, **D.J. Rasmussen**, F.M. Calafat, S. Vitousek, S. Dangendorf, R.E. Kopp, M. Oppenheimer. Observations reveal changing coastal storms around the United States. *Nature Climate Change*. doi:10.1038/s41558-025-02315-z [pdf]

2023 T.H.J. Hermans, V. Malagón-Santos, C.A. Katsman, R.A. Jane, **D.J. Rasmussen**, M. Haasnoot, G.G. Garner, R.E. Kopp, M. Oppenheimer, and A.B.A. Slangen. The Timing of Decreasing Coastal Flood Protection Due to Sea-Level Rise. *Nature Climate Change*. doi:10.1038/s41558-023-01616-5 [pdf]

2023 **D.J. Rasmussen**, R.E. Kopp, and M. Oppenheimer. Coastal flood protection megaprojects in an era of sea-level rise: politically feasible strategies or Army Corps fantasies? (Editor's Choice Collection) *Journal of Water Resources Planning and Management*. **149** (2) 10.1061/(ASCE)WR.1943-5452.0001613 [pdf]

2022 **D.J. Rasmussen**, S. Kulp, R.E. Kopp, M. Oppenheimer, and B. Strauss. Popular extreme sea level metrics can better communicate impacts (Opinion). *Climatic Change.* **170** (30) doi:10.1007/s10584-021-03288-6 [pdf]

2021 B. Strauss, S. Kulp, A. Levermann, and **D.J. Rasmussen**. Unprecedented threats to cities from multi-century sea level rise. *Environmental Research Letters*. **16** 114015 doi:10.1088/1748-9326/ac2e6b [pdf]

2021 C. Tebaldi, R. Ranasinghe, M. Vousdoukas, **D.J. Rasmussen**, B. Vega-Westhoff, E. Kirezci, R.E. Kopp, R. Sriver, and L. Mentaschi. Extreme Sea Levels at Different Global Warming Levels. *Nature Climate Change*. doi:10.1038/s41558-021-01127-1 [pdf]

2021 **D.J. Rasmussen**, R.E. Kopp, M. Oppenheimer, and R. Shwom. The political complexity of coastal flood risk reduction: lessons for climate adaptation public works in the U.S. *Earth's Future*. **9** (2) doi:10.1029/2020EF001575 [pdf]

2020 **D.J. Rasmussen**, M.K. Buchanan, R.E. Kopp, and M. Oppenheimer. A flood damage allowance framework for coastal protection with deep uncertainty in sea-level rise. *Earth's Future*. **8** (3) doi:10.1029/2019EF001340 [pdf]

2020 T. Frederikse, M.K. Buchanan, E. Lambert, R.E. Kopp, M. Oppenheimer, **D.J. Rasmussen**, and R.S.W. van de Wal. Antarctic Ice Sheet and emission scenario controls on 21st-century extreme sea-level changes. *Nature Communications*. **11** (390) doi:10.1038/s41467-019-14049-6 [pdf]

2018 **D.J. Rasmussen**, K. Bittermann, M.K. Buchanan, S. Kulp, B.H. Strauss, R.E. Kopp, and M. Oppenheimer. Extreme sea level implications of 1.5 °C, 2.0 °C, and 2.5 °C temperature stabilization targets in the 21st and 22nd century. *Environmental Research Letters*. **13** 034040 doi:10.1088/1748-9326/aaac87 [pdf]

2017 S.M. Hsiang, R.E. Kopp, A.S. Jina, J. Rising, M. Delgado, S. Mohan, **D.J. Rasmussen**, R. Muir-Wood, P. Wilson, M. Oppenheimer, K. Larsen, and T. Houser. Economic damage from climate change in the United States. *Science*. **356** (6345), 1362-1369 doi:10.1126/science.aal4369 [pdf]

2016 **D.J. Rasmussen**, M. Meinshausen, and R.E. Kopp. Probability-weighted ensembles of U.S. county-level climate projections for climate risk analysis. *Journal of Applied Meteorology and Climatology*. **55** (10), 2301-2322 doi:10.1175/JAMC-D-15-0302.1 [pdf]

2014 R.E. Kopp, R. Horton, C.M. Little, J.X. Mitrovica, M. Oppenheimer, **D.J. Rasmussen**, B.H. Strauss, and C. Tebaldi. Probabilistic 21st and 22nd century sea-level rise projections at a global network of tide gauge sites. *Earth's Future*. **2** (8), 383-406 doi:10.1002/2014EF000239 [pdf]

2013 **D.J. Rasmussen**, J. Hu, A. Mahmud, and M.J. Kleeman. The ozone climate penalty: past, present, and future. *Environmental Science & Technology*. **47** (24), 14258-14266 doi:10.1021/es403446m [pdf]

2012 **D.J. Rasmussen**, A.M. Fiore, V. Naik, L.W. Horowitz, M.G. Schultz, and S.J. McGinnis. Surface ozone-temperature relationships in the eastern US: A monthly climatology for evaluating chemistry-climate models. *Atmospheric Environment*. **47**, 142-153 doi:10.1016/j.atmosenv.2011.11.021 [pdf]

2011 **D.J. Rasmussen**, T. Holloway, and G.F. Nemet. Opportunities and Challenges in Assessing Climate Change Impacts on Wind Energy—A Critical Comparison of Wind Speed Projections in California. *Environmental Research Letters*. **6** 024008 doi:10.1088/1748-9326/6/2/024008 [pdf]

BOOKS

2015 T. Houser, R. Kopp, S. Hsiang, R. Muir-Wood, K. Larsen, **D.J. Rasmussen**, M. Delgado, A. Jina, S. Mohan, P. Wilson, M. Mastrandrea, and J. Rising: *Economic Risks of Climate Change: An American Prospectus.* New York, NY: Columbia University Press. 384 pp. ISBN: 9780231174565 [link]

SOFTWARE & DATA PRODUCTS

2026 **D.J. Rasmussen**. *SCOPE-ERA5: Station-Calibrated Outputs for Planning & Engineering—ERA5* [Data set]. Zenodo. doi:10.5281/zenodo.15611601 [link]

RESEARCH REPORTS

2015 U. Nopmongcol, J. Johnson, B. Brashers, **D.J. Rasmussen**, T. Shah, J. Zagunis, Z. Liu, and R. Morris: *Formation of Secondary $PM_{2.5}$ in the Capital Region Study.* Prepared for Environmental and Sustainable Resource Development, Edmonton, AB. [pdf]

2015 C. Emery, G. Wilson, **D.J. Rasmussen**, G. Yarwood: *CAMx Speed Improvements.* Prepared for Texas Commission on Environmental Quality, Austin, TX. [pdf]

2015 J. Johnson, G. Wilson, **D.J. Rasmussen**, G. Yarwood: *Daily Near Real-Time Ozone Modeling for Texas.* Prepared for Texas Commission on Environmental Quality, Austin, TX. [pdf]

2014 S. Kemball-Cook, T. Pavlovic, J. Johnson, L. Parker, **D.J. Rasmussen**, J. Zagunis, L. Ma, G. Yarwood: *Analysis of Wildfire Impacts on High Ozone Days in Houston, Beaumont, and Dallas-Fort Worth During 2012 and 2013.* Prepared for Texas Commission on Environmental Quality, Austin, TX. [pdf]

2014 T. Houser, R. Kopp, S. Hsiang, R. Muir-Wood, K. Larsen, **D.J. Rasmussen**, M. Delgado, A. Jina, S. Mohan, P. Wilson, M. Mastrandrea, and J. Rising: *American Climate Prospectus: Economic Risks in the United States.* Oakland, CA: Rhodium Group, 194 pp. [pdf]

2012 **D.J. Rasmussen** and M.J. Kleeman: Acceleration of CALPUFF using MPICH-2. Final Report to the California Air Resources Board. ARB Contract 11-760. [pdf]

2010 **D.J. Rasmussen**, L.M. Keller, and M.A. Lazzara: A 20-year assessment of the frequency and intensity of McMurdo area high wind events. Preprints, *5^{th} Antarctic Meteorological Observation, Modeling, and Forecasting Workshop* [pdf]

PRESENTATIONS

2026 ASHRAE Annual Conference, Austin, TX. “Global 1-km Climatic Design Extremes at Ungauged Locations via Machine Learning Regionalization of Extreme Value Parameters,” Paper Session 20: Advanced Heat Pumps, Ventilation, and Climate Data (*Poster*, AU-26-C074)

2023 American Geophysical Union Fall Meeting, San Francisco, CA. “Storm surge reanalysis of Hurricane Idalia using a spatial Bayesian model” (NH21D-06)

2023 American Geophysical Union Fall Meeting, San Francisco, CA. “Applying Allowances: A non-stationary, risk-based approach for advancing facility design to sea level rise and other climate hazards” (NH44A-02)

2021 NASA Goddard Institute for Space Studies (*Invited*)

2021 Woodwell Climate Research Center, Virtual (*Invited*)

2021 American Geophysical Union Natural Hazards Meeting, Virtual (*Invited*)

2021 Association of State Floodplain Managers Conference, Virtual

2021 UC-Irvine Departmental Seminar, Virtual (*Invited*)

2021 Coastal State Discussion Series (Rhode Island): The Future of Coastal Megaprojects, Virtual (*Invited*) [video]

2020 American Geophysical Union Fall Meeting, Virtual

2017 American Geophysical Union Fall Meeting, New Orleans, LA

2015 American Geophysical Union Fall Meeting, San Francisco, CA [poster]

2014 American Geophysical Union Fall Meeting, San Francisco, CA [slides]

2011 American Geophysical Union Fall Meeting, San Francisco, CA [slides]

PUBLIC OUTREACH

2021 The largest movable structures on Earth could save New York City from another Hurricane Sandy, but will they get built?, *Princeton Research Day* [Link]

2020 What can climate adaptation learn from what’s in Grandpa’s garage? A historical tale of two flood protection megastructures, *Highwire Earth* [Link]

2020 *Could megastructures save New York City from another Hurricane Sandy?*, short film (independently produced) [Link]

2020 Eight years after Hurricane Sandy New Yorkers are still waiting for flood protection, *C-PREE Blog* [Link]

AWARDS AND GRANTS

2020–2021 Dissertation Completion Fellowship/Post-graduate Research Associate (Princeton)

2019 Karl F. Schlaepfer ’49 and Gloria G. Schlaepfer Fund (Princeton)

PROFESSIONAL SERVICE

- American Society of Civil Engineers, Committee on Adaptation to a Changing Climate (2025–)
- U.S. Green Buildings Council, LEED v5 Resilience Working Group (2023–2024)
- Invited panelist for “Water, Water Everywhere: Climate Change, Flooding and the Gowanus Rezone”, Hosted by: *Voice of Gowanus*, 2021 [Link]
- Informal Reviewer, IPCC Special Report on the Ocean and Cryosphere in a Changing Climate, 2019
- American Meteorological Society, Energy Committee (2011–2013)
- Reviewer: *Nature Climate Change*, *Journal of Geophysical Research: Oceans*, *Environmental Research Letters*, *Earth's Future*, *Climatic Change*, *Scientific Reports*, *Atmospheric Chemistry and Physics*, *Atmospheric Environment*, *Journal of Flood Risk Management*, *Atmospheric Research*, *Journal of Waterway, Port, Coastal, and Ocean Engineering*, *Environmental Research Communications*, *Ocean & Coastal Management*, *Communications Earth & Environment*

TEACHING

TA, *The Environment: Science and Policy* (Princeton University, Spring 2017, 2018)

ADDITIONAL INFORMATION

Hobbies: Alpine skiing, CrossFit, AcroYoga, guitar, photography and film
Citizenship: USA

Last updated: July 9, 2026